NPTEL Online Certification Courses

Indian Institute of Technology Kharagpur

E-Business

Assignment- Week 9

TYPE OF QUESTION: MCQ/MSQ

Number of questions: 10 **Total mark: 10 X 2 = 20**

Question 1

A retail manager wants to evaluate the impact of a 20% increase in holiday-season demand on inventory levels before implementing any changes. Which DSS capability is most relevant?

a) Maintaining historical transaction records for future auditing purposes
b) Generating standardized reports summarizing previous inventory performance
c) Monitoring website activity using predefined operational dashboards
d) **Using analytical models to evaluate alternative scenarios before making decisions**

Correct Option: d
Explanation: DSS supports non-routine decision making by allowing managers to evaluate hypothetical scenarios using analytical models.
Week 9: Slide 4

Question 2

A business analyst develops a model that predicts customer demand while accounting for random fluctuations caused by weather and market conditions. This model is best described as:

a) A deterministic model where outcomes are fixed once inputs are specified
b) A static model where system behaviour remains unchanged over time
c) A descriptive model used only for summarizing historical observations
d) **A stochastic model that explicitly incorporates uncertainty into outcomes**

Correct Option: d
Explanation: Stochastic models incorporate randomness and uncertainty in the modelling process.
Week 9: Slide 6

Question 3

A company is interested in discovering frequently occurring phrases and themes from thousands of customer reviews. Which analytical approach is most suitable?

a) Time-series forecasting to estimate future review volumes
b) Decision analysis to compare alternative product strategies
c) Simulation modelling to generate synthetic customer opinions
d) **Text mining using natural language processing to identify patterns**

Correct Option: d
Explanation: Text mining employs natural language processing to extract patterns from textual data.
Week 9: Slide 9

Question 4

In an extended web log, which field would be most useful for determining whether a visitor arrived from a search engine or another website?

a) A field identifying the browser version used during the request
b) A field specifying the number of bytes transferred to the client
c) A field recording the HTTP method associated with the request
d) **A field containing the URL of the previously visited webpage**

Correct Option: d
Explanation: The Referrer field indicates the webpage from which the visitor arrived.
Week 9: Slide 22

Question 5

A company merges access logs from multiple web servers located in different regions. Which preprocessing activity is most critical before analyzing customer behaviour?

a) Generating association rules from the combined clickstream records
b) Calculating transition probabilities for all observed navigation paths
c) Estimating customer purchase intentions using predictive models
d) **Synchronizing and integrating data collected from different server logs**

Correct Option: d
Explanation: Data integration aligns information from multiple servers before analysis.
Week 9: Slide 31

Question 6

During web usage mining, a page view is best understood as:

a) A sequence of user visits occurring within a predefined timeout interval
b) A collection of customer transactions associated with a particular purchase
c) A group of sessions aggregated for multidimensional reporting purposes
d) **A set of page files and related objects contributing to one browser display**

Correct Option: d
Explanation: A page view includes the page and associated objects needed to display it.
Week 9: Slide 32

Question 7

A website experiences extensive caching, causing some user requests to bypass the server log. Which preprocessing activity specifically addresses this issue?

a) Removing requests associated with known search-engine crawlers
b) Grouping page requests into visits using timeout thresholds
c) Mapping URLs into higher-level semantic categories for analysis
d) **Inferring missing navigation steps through path completion techniques**

Correct Option: d
Explanation: Path completion reconstructs missing references caused by caching.
Week 9: Slide 37

Question 8

A retailer transforms URLs such as `/electronics/mobile/item123` and `/electronics/mobile/item456` into the concept "Mobile Phones." What is the primary advantage of this transformation?

a) It reduces the need for cleaning erroneous log entries from server records
b) It eliminates the requirement to identify sessions before mining begins
c) It ensures that all customers follow identical browsing sequences
d) **It allows navigation analysis to focus on concepts rather than specific URLs**

Correct Option: d
Explanation: Semantic modelling improves interpretation by mapping pages to business concepts.
Week 9: Slides 38–39

Question 9

A website records 4,000 hits during a day but only 800 page views. Which interpretation is most reasonable?

a) Every page view generated exactly five user sessions on average
b) Most requests originated from robots rather than human visitors
c) The website received more visits than page views during the period
d) **Many requests were for embedded objects such as images or scripts**

Correct Option: d
Explanation: Hits count all file requests, including images and scripts, whereas page views count pages.
Week 9: Slide 51

Question 10

An online bookstore discovers that customers frequently visit the “Programming Books” page and the “Machine Learning Books” page during the same session. Which mining technique is most likely responsible for uncovering this relationship?

a) Classification methods assigning users to predefined reader categories
b) Clustering approaches grouping visitors with similar demographics
c) Markov-chain models estimating probabilities of future transitions
d) **Association-rule mining identifying frequently co-occurring items**

Correct Option: d
Explanation: Association rules identify items or pages that frequently occur together.
Week 9: Slide 58

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